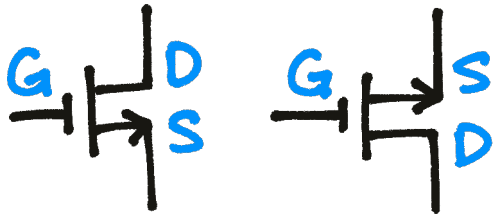
NMOS

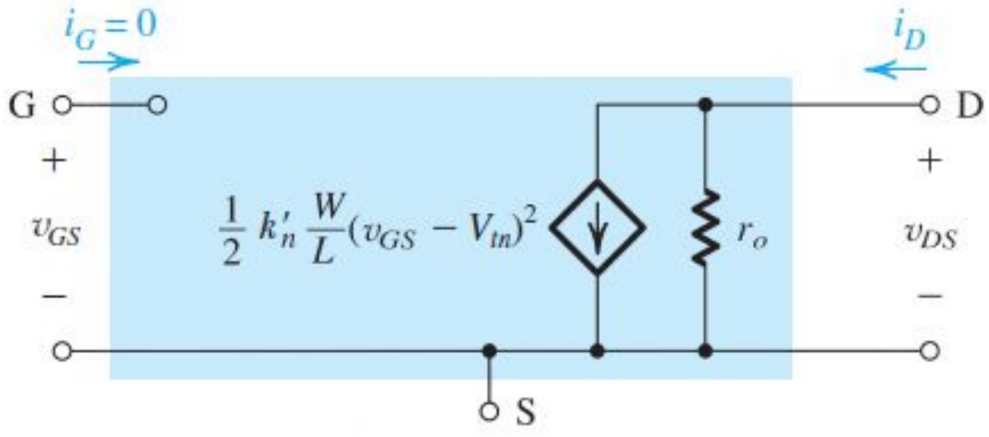
Schematic

Symbols

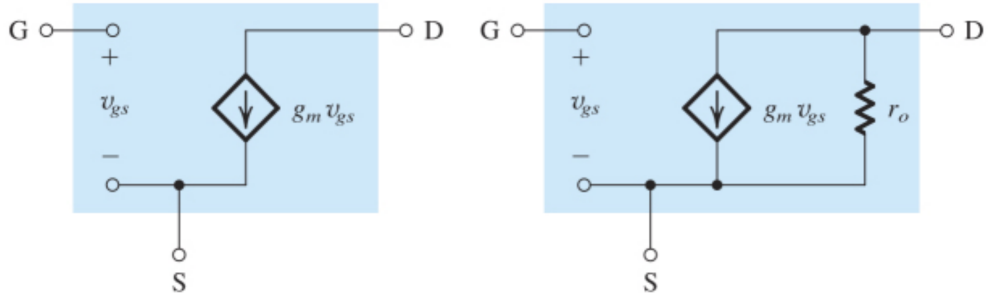


Models

Large-signal model:



Small-signal model:



Equations

Equation	Conditions	Notes
$i_D = 0$	$v_{GS} < V_{tn}$	(Cutoff region)
$i_D = k'_n \left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right) v_{DS}$	$v_{DS} < v_{OV}$	(Triode region)
$i_D = \frac{1}{2}k'_n \left(\frac{W}{L}\right) v_{OV}^2 \left(1 + \lambda v_{DS}\right)$	$v_{DS} \geq v_{OV}$	(Saturation region)
$k'_n = \mu_n C_{ox}$		
$v_{OV} = v_{GS} - V_t$		
$\lambda = \frac{1}{V_A}$		
$V_A = V'_A L$		
$k_n = k'_n \left(\frac{W}{L}\right)$		
$g_{DS} = k_n v_{OV}$		
$r_{DS} = \frac{1}{g_{DS}}$		
$A_v \equiv \frac{v_{ds}}{v_{gs}} = -g_m R_D$		
$g_m \equiv \frac{\partial i_D}{\partial v_{GS}}$	$v_{GS} = V_{GS}$	g_m is slope at bias point
$g_m = \sqrt{2k_n I_D}$		
$g_m = \frac{2I_D}{V_{OV}}$		
$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1}$	v_{GS} constant	
$r_o = \frac{V_A}{i'_D}$		
$i'_D = \frac{1}{2}k'_n \left(\frac{W}{L}\right)(v_{GS} - V_t)^2$		

Notes

Term	Description
k'_n	Process transconductance parameter
k_n	MOSFET transconductance parameter
i'_D	Drain current without channel-length modulation
μ_n	Electron mobility
C_{ox}	Oxide capacitance
L	Channel length
V'_A	Process-technology dependent parameter
A_v	Voltage gain
v_{OV}	Overvoltage
V_A	Early voltage
V_t	Threshold voltage
g_{DS}	Channel conductance
r_{DS}	Channel resistance
g_m	Mutual gain